



BANKING ANALYTICS PROJECT

Banking Customer & Risk Analytics

A plain-language walkthrough of customers, loans, transactions and branch performance

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Built with Python · MySQL · Power BI

What This Presentation Covers

No technical background needed — every number is explained in plain business terms.

1

The Problem

Why banks need one clear, trustworthy view of their data

2

The Solution

How raw records became one easy-to-use dashboard

3

The Dashboard

A look at the four report pages built for the business

4

Key Numbers

The headline figures every leader should know

5

What We Found

Insights about customers, loans, risk and branches

6

What To Do Next

Practical recommendations and the roadmap ahead

THE PROBLEM

Data Was Everywhere — Answers Were Nowhere

Every day, a bank creates huge amounts of data — new accounts, loan repayments, card swipes, UPI transfers. But this data usually sits scattered across separate systems, making simple questions surprisingly hard to answer quickly.

- Which customers are most valuable — and which might leave?
- Which loans are most likely to default, and why?
- Which branches and regions are performing best?
- Are customers financially comfortable, or stretched thin?



The Goal

Bring five separate banking datasets together into one clean, reliable source — and present it through a dashboard any manager can use, technical or not, to make faster, evidence-based decisions.

This project informs decisions — it does not automate lending decisions.

From Raw Records to a Ready-to-Use Dashboard

Three simple stages turn scattered spreadsheets into business answers.

1



Raw Data

5 linked datasets:
75K customers · 85K accounts
30K loans · 500K transactions
100 branches



2



Clean & Organise

Python fixes missing data,
removes duplicates, and
creates
business fields like income
group
and risk category



3



Analyse

MySQL answers 35+ real
business questions, from
top branches to default rates
by loan type



4



Present

Power BI turns the results
into one interactive dashboard
—
filter and explore with no
coding required

KEY NUMBERS

The Bank, At a Glance

These are the headline figures every leader should know.

75,000

Total Customers

₹6bn

Total Transaction Value

₹2.41bn

Total Deposits

30,000

Total Loans

5.39%

Loan Default Rate

500K

Total Transactions

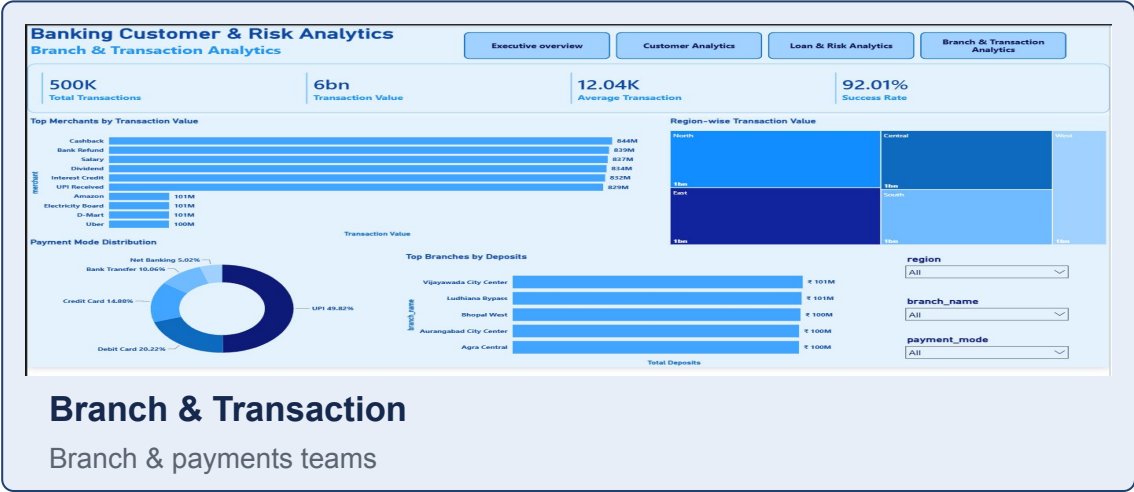
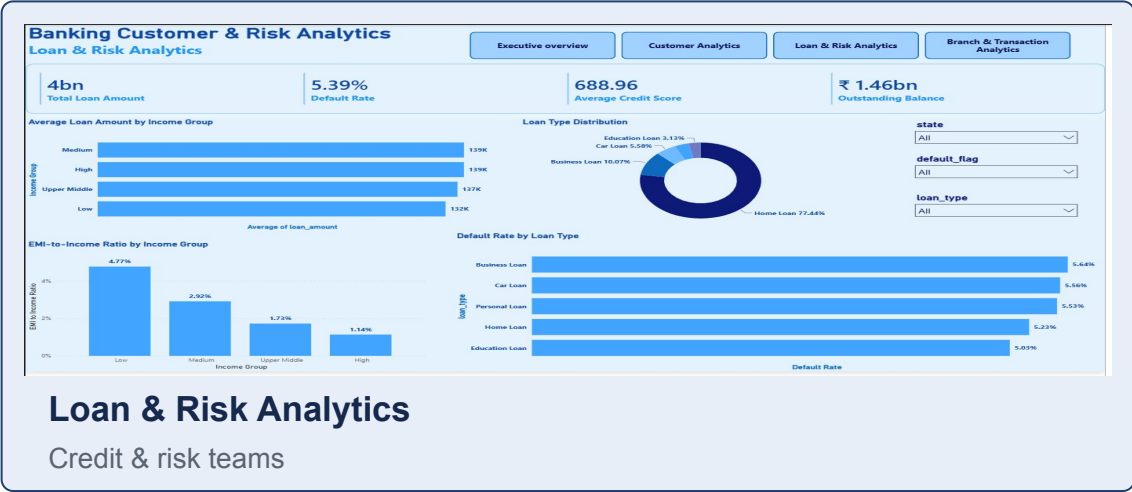
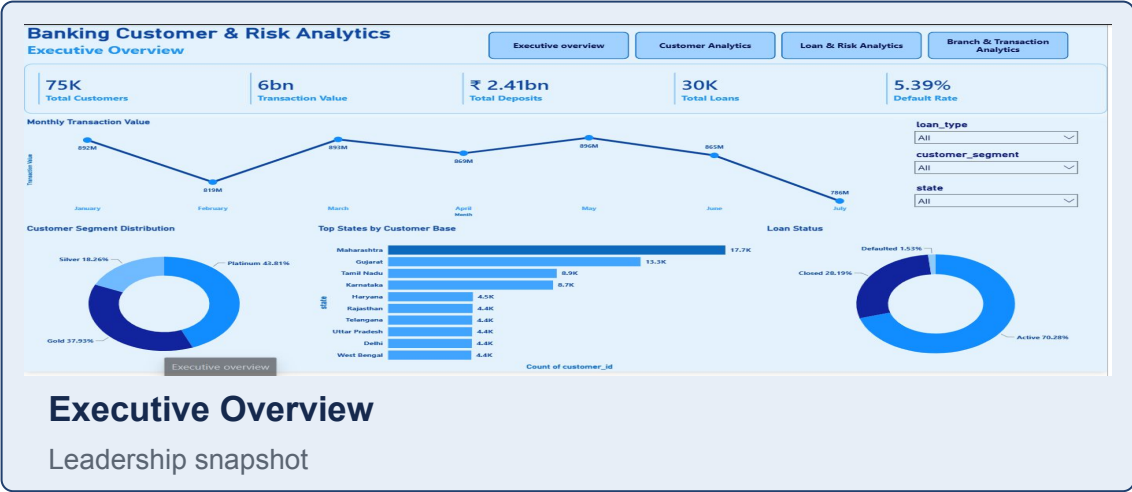
92.01%

Transaction Success Rate

₹4bn

Total Loan Amount

One Dashboard, Four Views — Built for Every Team



Who Our Customers Are



Balanced Base

52% male, 48% female — a broadly even customer mix.



Middle-Aged Majority

Customers aged 30-59 make up roughly two-thirds of the base.



Mostly Premium

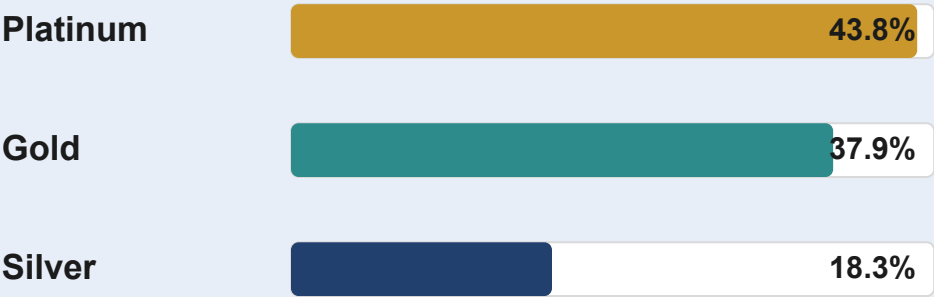
Platinum (43.8%) is the largest customer tier — a broadly loyal, affluent base.



A Few Key States

Maharashtra, Gujarat, Tamil Nadu & Karnataka hold ~half of all customers.

Customer Segment Mix

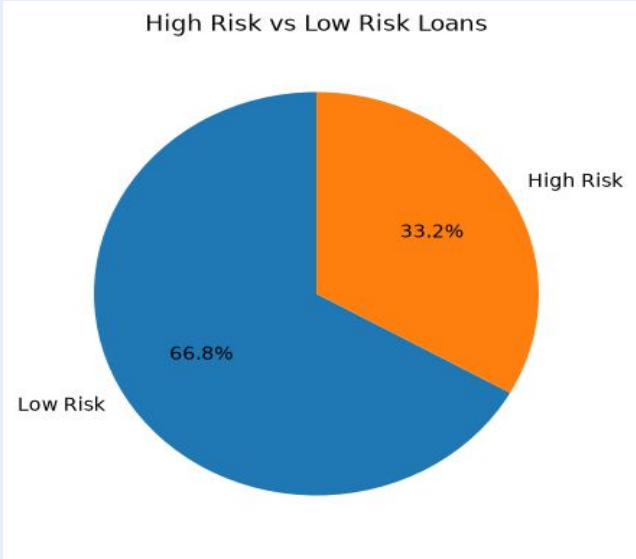


Top States by Customer Base

- Maharashtra — 17.7K
- Gujarat — 13.3K
- Tamil Nadu — 8.9K
- Karnataka — 8.7K

How Healthy Is the Loan Book?

Loan Risk Split



2 in 3 loans are Low Risk

(66.8% low risk vs 33.2% high risk)

Credit Score Is a Strong Predictor

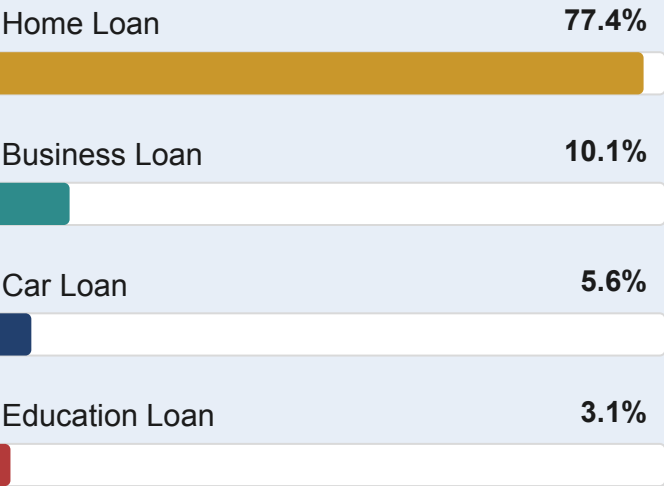
21.68%

default rate for "Poor" credit-score customers

5.39%

overall bank-wide default rate — roughly 4x lower

What Do People Borrow For?

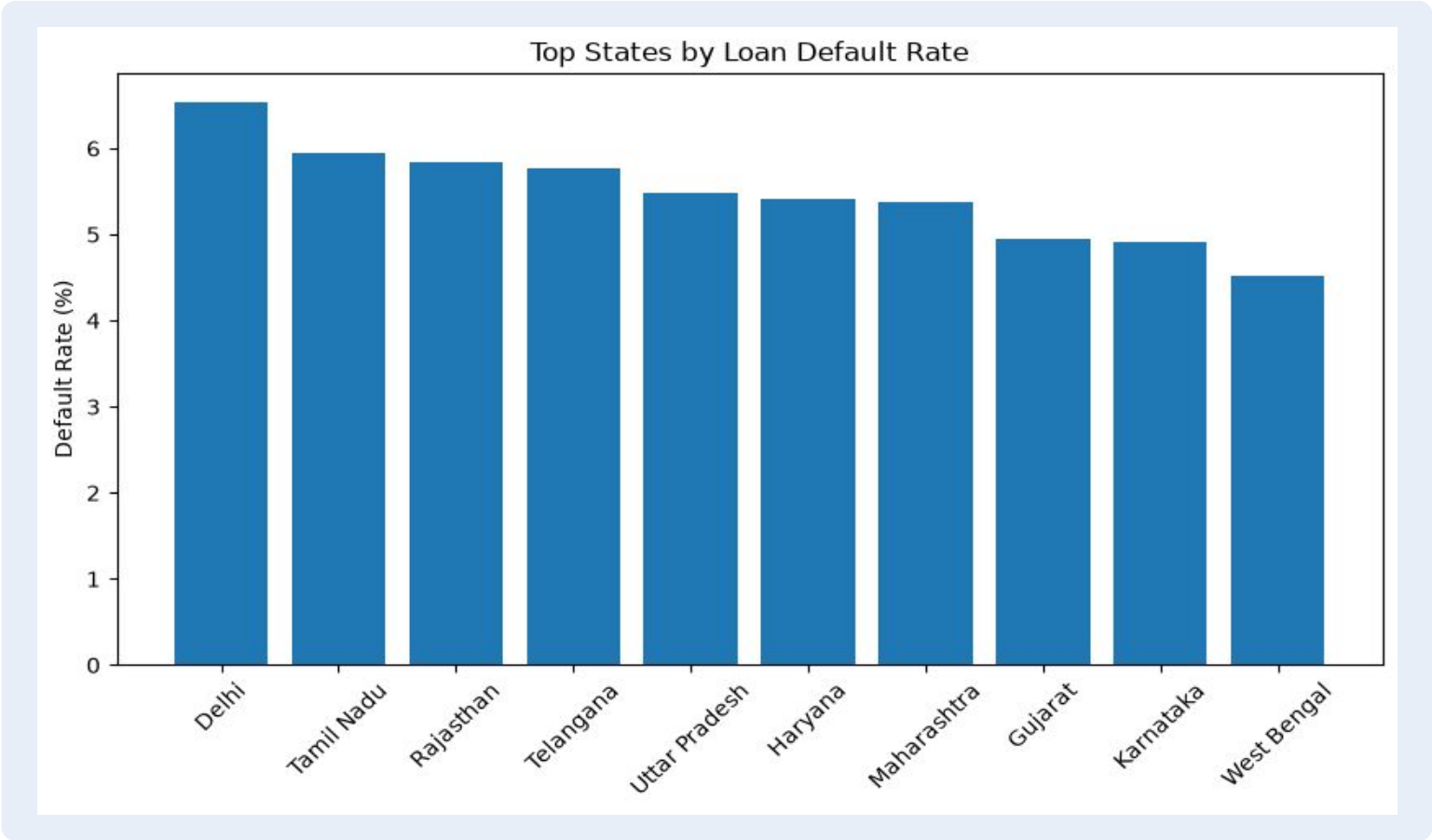


EMI Burden by Income

Low income: 4.77% of monthly income → High income: 1.14%

Where Loan Risk Is Highest

Default rate by state — states are not equally risky.



Highest Risk

Delhi — 6.5%
Tamil Nadu — 5.9%
Rajasthan — 5.8%

Lowest Risk

West Bengal — 4.5%
Karnataka — 4.9%
Gujarat — 4.95%

Useful for targeting collections support and regional lending policy.

How Money Moves Through the Bank

49.8%

of transactions are via UPI — the leading payment method

92.01%

transaction success rate — about 1 in 12 currently fail

₹100M+

held by each of the top 5 branches by deposits

₹0.75bn

West region transaction value — the lowest of all regions

Regional Transaction Value



Recommendations for the Business



Tighten lending for "Poor" scores

Add stricter checks or risk-based pricing — this group defaults 4x the average.



Support stretched borrowers early

Reach out to lower-income customers before repayment strain turns into default.



Fix the 8% failed transactions

Even small gains in success rate protect meaningful value at this transaction volume.



Learn from top-deposit branches

Study what Vijayawada, Ludhiana, Bhopal, Aurangabad & Agra do well, then replicate it.



Invest in the West region

Transaction value there trails every other region — a clear growth opportunity.



Use segments for targeted offers

Design retention offers by Platinum/Gold/Silver tier, and grow Silver into Gold.



Thank You

Questions and feedback are welcome.

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Full technical documentation available in the project report and README.